PYTHON ASSIGNMENT 02

Data Structures & Conditional Statements

Practical Implementation Report

STUDENT	Shanmukh Koyya
TOOLS	Python • Jupyter Notebook
FOCUS	Lists • Dictionaries • Sets • Conditional Statements

Prepared as a structured record of the completed Python Assignment 2 exercises.

01 | Project Overview

This assignment focuses on fundamental Python data structures and conditional statements. The exercises provide hands-on practice with Lists, Dictionaries, Sets, and score-based conditional logic.

Objectives

• Create and modify Python lists.

• Perform common list operations and modifications.

• Access list elements using indexing and slicing.

• Create and manipulate dictionaries.

• Work with dictionary keys, values, and items.

• Understand duplicate removal and operations in sets.

• Apply conditional statements using user input.

02 | Lists

The assignment begins with age_list containing five integer elements and name_list containing five string elements.

List Operations & Modifications

Operation	Python method / function
Append a name	name_list.append("Yazhini")
Insert an age	age_list.insert(2, 30)
Remove a name	name_list.remove("Yazhini")
Pop last age	age_list.pop()
Extend the age list	age_list.extend([29, 30, 26])
Sort descending	age_list.sort(reverse=True)
Maximum / Minimum / Sum	max(age_list), min(age_list), sum(age_list)

Accessing List Elements

Requirement	Example
First element	name_list[0]
Last element	name_list[-1]
Elements at indexes 2 to 4	name_list[2:5]
Reverse the list	name_list.reverse()

Example

```
age_list = [24, 25, 26, 27, 28]
name_list = ["Ram", "Raj", "Rahul", "Rakesh", "Rajesh"]
name_list.append("Yazhini")
```

```
age_list.insert(2, 30)
age_list.extend([29, 30, 26])
age_list.sort(reverse=True)
```

03 | Dictionary

A student_marks dictionary stores student names and their marks. The assignment demonstrates accessing, adding, updating, and inspecting dictionary data.

Dictionary Tasks

Task	Example
Create dictionary	student_marks = {"Ram": 85, ...}
Access a student's mark	student_marks["Rahul"]
Add Janani	student_marks["Janani"] = 80
Update an existing mark	student_marks["Raj"] = 82
Print keys	student_marks.keys()
Print values	student_marks.values()
Print items	student_marks.items()

Example

```
student_marks = {
"Ram": 85,
"Raj": 82,
"Rahul": 92,
"Rakesh": 74,
"Rajesh": 88
}
student_marks["Janani"] = 80
```

04 | Sets

The set exercises demonstrate duplicate removal and the fact that set elements cannot be accessed through indexed item assignment. Two sets are then used to practice union and intersection.

Operation	Result
Duplicate removal	Repeated a and i occur only once in the set
Index assignment	my_set[4] = 's' produces a TypeError
Union	{1, 2, 3, 5, 7, 8, 9, 10}
Intersection	{3, 5}

```
set1 = {1, 3, 5, 7, 9}
set2 = {2, 3, 5, 8, 10}
print(set1.union(set2))
print(set1.intersection(set2)
```

05 | Conditional Statements

The final section accepts a score from 0 to 10 and assigns a performance category according to the supplied conditions.

Score condition	Category
Score > 7	Above Average
Score 4–7	Average
Score < 4	Below Average

Implementation

```
score = int(input("Enter your score (0 to 10): "))
if score > 7:
print("Above Average: Excellent performance! Keep up the good work.")
elif score >= 4:
print("Average: Good effort! Keep practicing, there is room for improvement.")
else:
print("Below Average: Need to improve your performance. Consistent practice will lead
to better results.")
```

Sample Output

Enter your score (0 to 10): 7
Average: Good effort! Keep practicing, there's room for improvement.

06 | Key Learning Outcomes

• Practiced Python list creation, modification, indexing, slicing, and built-in functions.

• Worked with dictionary key-value pairs and dictionary methods.

• Observed how sets remove duplicate values and support set operations.

• Applied if, elif, and else statements to classify user input.

• Strengthened basic Python programming logic through practical exercises.

07 | Conclusion

This assignment provided practical practice with core Python data structures and conditional statements. The completed exercises demonstrate the use of lists, dictionaries, sets, indexing, built-in functions, and conditional logic in a structured programming workflow.

Author

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